Product Backlog: Step Sequencer

CBL Project

Group members: Andria Jakobia (2416220), Aadidev Anuroop (2442566)

Product description

A step sequencer is a simple drum machine. The user sees a grid: each row is a drum sound (kick, snare, hi-hat, clap) and each column is one beat-step in a loop. The user clicks cells to turn them on and presses play. A highlight then moves across the grid from left to right, playing the sounds in the active cells. This builds a repeating drumbeat, as in music production software. The tempo (speed) is adjustable, and the finished beat can be saved to a file and loaded again later.

Glossary

- **BPM:** beats per minute, i.e. the playback speed.
- **Velocity:** how loud a single step is played.
- **Sample:** a short, recorded sound file, e.g. one kick drum hit.

Must have

#	Name	How to demo	Notes
1	Step grid	1. Launch the app. 2. A window opens with 4 labelled rows (Kick, Snare, Hi-hat, Clap), each with 16 empty cells.	Swing panel with a 4x16 grid layout. Which cells are on is stored in a separate pattern class (a 4x16 table of true/false), not in the buttons.
2	Toggle cells	1. Click an empty cell: it changes color. 2. Click it again: it goes back to empty.	Each click flips the matching true/false value in the pattern class and repaints that cell.
3	Play and stop	1. Press Play: playback starts. 2. Press Stop: playback stops and the position resets to step 1.	Playback runs on its own thread, so the window does not freeze. Pressing Play twice must not start two threads.
4	Playback of sounds	1. Turn on the Kick cells at steps 1, 5, 9 and 13. 2. Press Play: a kick is heard four times per loop, evenly spaced.	Loop: for each step, play every row that is on, then wait one step length. Sound clips are loaded once at start-up and bundled in the project, so it runs on Windows, macOS and Linux.
5	Playhead highlight	1. Press Play. 2. A highlight moves along the columns one step at a time, in time with the sound.	The playback thread tells the GUI which step is current. GUI updates must run on the Swing thread
6	Tempo slider	1. Press Play. 2. Move the BPM slider right: the loop speeds up. Move it left: it slows down.	Step length = 15000 / BPM milliseconds (16th notes). The slider is read every step, so changes apply immediately.

Should have

#	Name	How to demo	Notes
7	Sound selection per row	1. Open the dropdown on the Kick row and choose another kick. 2. Press Play: the new sound is heard.	3-4 sample files per instrument; the dropdown swaps which clip the row uses.
8	Clear pattern	1. Turn on several cells. 2. Press Clear: all cells are empty.	Sets every value in the pattern to false and repaints the grid.

Advanced topics

1. Audio playback with Java Sound
2. Multithreading